

Department of Computer Science**Assignment 4****Course Code: CS-151****Name:** _____**Course Title: Database Systems****Registration No.** _____**Total Marks: 20****Discipline/Semester: BSCS 2nd****Date: 11-06-2025****Submission Date: 20-06-2025****Instructions:**

- **Late submissions will not be accepted.**
- **Copied assignments from any source (including AI) will be marked zero.**

Question: 1 [GA4, CLO3, C4] (10 Marks)

Write SQL queries for the following tasks using a table named PRODUCT. First, create and populate the table, then answer each query question below.

i. Create the PRODUCT table with the following fields and insert data for at least 15 products:

- product_id
- product_name
- category
- supplier_id
- manufacture_date
- price
- discount
- warehouse_id

ii. Write an SQL query to find the prices of all products.

iii. Write an SQL query to find the unique product categories, displaying them as "Category".

iv. Write a SQL quer to list each product's name and increase its price by 20%, displayed as "Discounted Price (\$)".

v. Write a SQL qury to list product name and category in the format: "Product: NAME - Category: CATEGORY".

vi. Write an SQL query to produce output like: Product LAPTOP (electronics)

- vii. Write an SQL query to list products with manufacture dates formatted as "JANUARY 10, 2023". Display product ID, product name, price, and formatted manufacture date.
- viii. Write an SQL query to find all unique combinations of warehouse and category. Return warehouse ID and category.
- ix. Write an SQL query to find all products in the 'GADGETS' category. Return full product details.
- x. Write an SQL query to list all products with a price less than \$500. Return full product details.

Question: 2 (10 marks)

Consider the following table, Identify functional dependencies and discuss database anomalies in this table. Resolve the anomalies by normalizing database tables upto 4NF.

Project Code	Project Name	Project Manager	Project Budget	Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
PC010	Reservation System	Mr. Ajay	120500	S100	Mohan	D03	Database	21.00
PC010	Reservation System	Mr. Ajay	120500	S101	Vipul	D02	Testing	16.50
PC010	Reservation System	Mr. Ajay	120500	S102	Riyaz	D01	IT	22.00
PC011	HR System	Mrs. Charu	500500	S103	Pavan	D03	Database	18.50
PC011	HR System	Mrs. Charu	500500	S104	Jitendra	D02	Testing	17.00
PC011	HR System	Mrs. Charu	500500	S315	Pooja	D01	IT	23.50
PC012	Attendance System	Mr. Rajesh	710700	S137	Rahul	D03	Database	21.50
PC012	Attendance System	Mr. Rajesh	710700	S218	Avneesh	D02	Testing	15.50
PC012	Attendance System	Mr. Rajesh	710700	S109	Vikas	D01	IT	20.50